Multi-Agent Scientific Paper Review

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File	Final_paper_kjns_210426 (1).pdf
Words	12,275
Sections Detected	preamble, introduction, methodology, experiments, discussion, section_vi, conclusions, section_vii, biogra
Review Mode	FAST
Model	qwen3:8b

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Editorial Decision

MINOR REVISION

Weighted Score: 3.6 / 5.0
Confidence: 83%

Evaluation Score Table

Scale: 0 = Not evaluable | 1 = Very weak | 2 = Weak | 3 = Acceptable | 4 = Good | 5 = Excellent

Criterion	Score	Rating	Visual
Originality & Novelty	**5.0**	Excellent	■■■■■
Manuscript Structure	**2.5**	Weak	■■■■■
Methodology & Equations	**4.0**	Good	■■■■■
Statistical Analysis	**3.5**	Good	■■■■■
Results & Evidence	**4.0**	Good	■■■■■
Discussion & Conclusions	**3.5**	Good	■■■■■
Figures & Tables	**4.5**	Excellent	■■■■■
Scientific Writing	**3.5**	Good	■■■■■
References & Citations	**2.5**	Weak	■■■■■
Ethics & Transparency	**1.5**	Very Weak	■■■■■

Detailed Agent Reviews

TitleAbstractKeywordsReviewer

Title, Abstract, Keywords

Score: **5.0 / 5.0** | Confidence: 95%

Strengths:

- [TITLE] is specific, informative, and includes key terms for discoverability.
- [ABSTRACT] follows a structured format with clear background, objective, methods, results, and conclusions.
- [KEYWORDS] include a mix of specific and general terms, with 8 keywords that are relevant to the domain.

Minor Comments:

- [TITLE] could be slightly shortened for clarity without losing specificity, but it is already within acceptable length.

Recommendations:

- [TITLE] could consider removing 'via' and 'under' for conciseness while maintaining clarity.

StructureReviewer

Manuscript structure and logical coherence

Score: **2.5 / 5.0** | Confidence: 80%

Strengths:

- Clear section mapping
- Logical flow between sections where content is present

Weaknesses:

- Missing literature review section
- Inconsistent content across sections (e.g., Section V and Section VI repeat content)
- Lack of clear hypothesis or research question
- Section balance is uneven with some sections being disproportionately long or absent

Major Comments:

- The manuscript is missing a critical literature review section, which is essential for contextualizing the research problem and existing work.
- The content in Sections V and VI is repetitive, which undermines the logical flow and coherence of the manuscript.
- The research problem and hypothesis are not clearly stated, which is a fundamental requirement for a scientific manuscript.

Minor Comments:

- Section titles are not consistently formatted or descriptive.
- Some sections contain incomplete or truncated content, which affects readability and coherence.

Recommendations:

- Include a comprehensive literature review section to contextualize the research problem and existing work.
- Revise Sections V and VI to remove redundancy and ensure logical progression.
- Clearly state the research problem, hypothesis, and objectives in the introduction and ensure they are addressed in the conclusions.
- Ensure all sections are balanced and contribute meaningfully to the overall narrative.

MethodologyReviewer

Methodology, experimental design, equations — holistic review

Score: **4.0 / 5.0** | Confidence: 90%

Strengths:

- Clear integration of physical constraints into the neural network framework
- Logical flow from problem definition to experimental validation
- Use of a multi-objective optimization approach for parameter identification

Weaknesses:

- Lack of detailed description of the exact evolutionary algorithm used (e.g., genetic algorithm vs. particle swarm optimization)
- Insufficient detail on the exact formulation of the multi-objective optimization problem
- No mention of the exact neural network architecture or training parameters

Major Comments:

- The manuscript should provide more detailed information on the specific evolutionary algorithm (e.g., genetic algorithm, particle swarm optimization) and its implementation.

Minor Comments:

- Clarify the exact formulation of the multi-objective optimization problem.

Recommendations:

- Include a detailed description of the neural network architecture, training parameters, and the exact implementation of the evolutionary algorithm.

StatisticsReviewer

Statistical methods, tests, and reporting quality

Score: **3.5 / 5.0** | Confidence: 80%

Strengths:

- Use of Kruskal-Wallis and post-hoc Wilcoxon/Holm corrections for non-parametric comparisons
- Bootstrap confidence intervals for robustness
- RMSE, MSE, and coefficient of variation for performance evaluation
- Pareto/hypervolume indicators for multi-objective optimization

Weaknesses:

- Lack of effect size reporting
- No verification of test assumptions
- No reproducibility details provided

Major Comments:

- The manuscript should verify the assumptions of the Kruskal-Wallis and Wilcoxon tests (e.g., independence, distribution type).
- Effect sizes (e.g., Cohen's d, eta-squared) should be reported alongside p-values to assess practical significance.
- Reproducibility details such as random seed values, number of resamples for bootstrap, and variance estimates should be included for transparency.

Minor Comments:

- Consider clarifying the interpretation of coefficient of variation and hypervolume indicators.
- Ensure that the 30 independent runs are justified in the context of the study's objectives and computational constraints.

Recommendations:

- Verify the assumptions of non-parametric tests (e.g., independence, distribution type) and report results accordingly.
- Report effect sizes to complement p-values and provide a more complete picture of the findings.
- Include reproducibility details such as random seeds, number of resamples for bootstrap, and variance estimates to ensure transparency and replicability.

FiguresTablesEquationsReviewer

Figures, tables, equations — quality and coherence with text

Score: **4.5 / 5.0** | Confidence: 80%

Strengths:

- The manuscript presents a comprehensive methodology and experimental validation framework.
- The use of sequential equations and table references suggests a well-organized structure.

Weaknesses:

- The extracted text does not expose table bodies, making it difficult to verify data consistency and completeness.

Major Comments:

- The manuscript should provide the full content of tables and figures to ensure data consistency and coherence with the text.

Minor Comments:

- Ensure that all figures and tables are fully described and self-explanatory, even if the body content is not visible.

Recommendations:

- Include the full content of tables and figures to support the claims made in the text.

ResultsReviewer

Results validity and empirical support

Score: **4.0 / 5.0** | Confidence: 80%

Strengths:

- MSE, RMSE, CV, and hypervolume metrics are reported, which are relevant to the multi-objective optimization problem
- Pareto analysis and baseline comparisons with GA and PSO are included
- 30 independent runs are mentioned, indicating robustness in the experimental design

Weaknesses:

- The results section is incomplete and cut off, making it difficult to assess the full extent of the quantitative results and their consistency with figures and tables
- Negative results are not reported, which could affect the completeness of the findings
- The section lacks a clear statement of how the results align with the stated research objectives

Major Comments:

- The results section is incomplete and cut off, which prevents a thorough evaluation of the quantitative results and their consistency with the figures and tables.

Minor Comments:

- The section lacks a clear statement of how the results align with the stated research objectives.

Recommendations:

- Complete the results section to ensure all quantitative results are fully presented and aligned with figures and tables.
- Include a discussion of negative results to provide a balanced view of the findings.

DiscussionConclusionsReviewer

Discussion depth and conclusions validity

Score: **3.5 / 5.0** | Confidence: 90%

Strengths:

- The discussion provides a clear comparison of the EANN approach with GA and PSO in terms of performance metrics and computational cost.
- The conclusions are directly tied to the experimental results and the research objectives.

Weaknesses:

- The discussion lacks a detailed comparison with prior work in the literature, which limits the contextualization of the findings.
- The discussion does not address unexpected findings or limitations of the study beyond computational cost.
- The conclusions do not specify future work directions, which is a key criterion for Q1 journals.

Major Comments:

- The manuscript should include a more comprehensive comparison with prior work to contextualize the findings and demonstrate the novelty of the EANN framework.

Minor Comments:

- The discussion could benefit from addressing any limitations of the study beyond computational cost.

Recommendations:

- Include a detailed comparison with existing literature to highlight the novelty and significance of the EANN framework.

WritingReviewer

Scientific writing quality, clarity, and style

Score: **3.5 / 5.0** | Confidence: 80%

Strengths:

- Clear technical terminology
- Logical structure of sections

Weaknesses:

- Overuse of passive voice
- Vague quantifiers
- Undefined acronyms

Major Comments:

- The manuscript contains several instances of passive voice that could be rephrased for clarity and directness.
- The use of 'significant' without numerical context weakens the scientific rigor.
- The acronym EANN is used without definition, which is a critical omission for clarity.

Minor Comments:

- Some sentences are overly complex and could be simplified for better readability.
- The introduction lacks a clear research gap or motivation for the proposed method.

Recommendations:

- Define all acronyms on first use.
- Avoid passive voice where active voice is possible.
- Provide numerical context for vague quantifiers like 'significant'.
- Simplify complex sentences for improved readability.

ReferencesReviewer

References quality, recency, and relevance

Score: **2.5 / 5.0** | Confidence: 80%

Strengths:

- Recent references from 2021-2023
- Consistent citation style (IEEE format)

Weaknesses:

- Lack of seminal foundational works
- Missing key areas in literature review

Major Comments:

- The references section lacks seminal foundational works that are critical for understanding the theoretical basis of lumped mass-spring-damper models and structural health monitoring. This omission weakens the manuscript's ability to establish a strong theoretical foundation.
- The manuscript does not include classic works that are foundational to the field, which is essential for a comprehensive literature review.

Minor Comments:

- The references section includes a mix of conference and journal papers, which is acceptable but could be balanced further for a more comprehensive review.

Recommendations:

- Include seminal foundational works on structural health monitoring and lumped mass-spring-damper models.
- Add classic references that are foundational to the field to strengthen the literature review.

EthicsLimitationsReviewer

Research ethics, limitations, and transparency

Score: **1.5 / 5.0** | Confidence: 80%

Strengths:

- The limitations are explicitly stated in the discussion section, indicating an awareness of the study's constraints.

Weaknesses:

- Limitations are not specific or detailed, which undermines the credibility of the study's self-assessment.
- Reproducibility is not addressed, which is critical for scientific transparency and validation.
- Data availability is not stated, which is essential for open science and reproducibility.
- Conflict of interest, funding sources, and author contributions are not disclosed, which is a major omission in ethical reporting.

Major Comments:

- The manuscript lacks essential ethical reporting elements such as ethical approval, data availability, conflict of interest, and funding disclosure. These are critical for transparency and reproducibility.

Minor Comments:

- The limitations are mentioned but not sufficiently detailed or specific.

Recommendations:

- Include a statement on ethical approval if applicable.
- Provide a data availability statement to ensure reproducibility.
- Declare any potential conflicts of interest and disclose funding sources.
- Elaborate on the limitations with specific examples and their impact on the study's conclusions.

Consolidated Major Comments

These issues must be addressed before the manuscript can be accepted.

[1] The manuscript is missing a critical literature review section, which is essential for contextualizing the research problem and existing work.

[2] The content in Sections V and VI is repetitive, which undermines the logical flow and coherence of the manuscript.

[3] The research problem and hypothesis are not clearly stated, which is a fundamental requirement for a scientific manuscript.

[4] The manuscript should provide more detailed information on the specific evolutionary algorithm (e.g., genetic algorithm, particle swarm optimization) and its implementation.

[5] The manuscript should verify the assumptions of the Kruskal-Wallis and Wilcoxon tests (e.g., independence, distribution type).

[6] Effect sizes (e.g., Cohen's d, eta-squared) should be reported alongside p-values to assess practical significance.

[7] Reproducibility details such as random seed values, number of resamples for bootstrap, and variance estimates should be included for transparency.

[8] The manuscript should provide the full content of tables and figures to ensure data consistency and coherence with the text.

[9] The results section is incomplete and cut off, which prevents a thorough evaluation of the quantitative results and their consistency with the figures and tables.

[10] The manuscript should include a more comprehensive comparison with prior work to contextualize the findings and demonstrate the novelty of the EANN framework.

[11] The manuscript contains several instances of passive voice that could be rephrased for clarity and directness.

[12] The use of 'significant' without numerical context weakens the scientific rigor.

Consolidated Minor Comments

These are suggestions that would improve the manuscript quality.

[1] [TITLE] could be slightly shortened for clarity without losing specificity, but it is already within acceptable length.

[2] Section titles are not consistently formatted or descriptive.

[3] Some sections contain incomplete or truncated content, which affects readability and coherence.

[4] Clarify the exact formulation of the multi-objective optimization problem.

[5] Consider clarifying the interpretation of coefficient of variation and hypervolume indicators.

[6] Ensure that the 30 independent runs are justified in the context of the study's objectives and computational constraints.

[7] Ensure that all figures and tables are fully described and self-explanatory, even if the body content is not visible.

[8] The section lacks a clear statement of how the results align with the stated research objectives.

[9] The discussion could benefit from addressing any limitations of the study beyond computational cost.

[10] Some sentences are overly complex and could be simplified for better readability.

[11] The introduction lacks a clear research gap or motivation for the proposed method.

[12] The references section includes a mix of conference and journal papers, which is acceptable but could be balanced further for a more comprehensive review.

Specific Improvement Recommendations

1. [TITLE] could consider removing 'via' and 'under' for conciseness while maintaining clarity.
2. Include a comprehensive literature review section to contextualize the research problem and existing work.

3. Revise Sections V and VI to remove redundancy and ensure logical progression.

4. Clearly state the research problem, hypothesis, and objectives in the introduction and ensure they are addressed in the conclusions.

5. Ensure all sections are balanced and contribute meaningfully to the overall narrative.

6. Include a detailed description of the neural network architecture, training parameters, and the exact implementation of the evolutionary algorithm.

7. Verify the assumptions of non-parametric tests (e.g., independence, distribution type) and report results accordingly.

8. Report effect sizes to complement p-values and provide a more complete picture of the findings.

9. Include reproducibility details such as random seeds, number of resamples for bootstrap, and variance estimates to ensure transparency and replicability.

10. Include the full content of tables and figures to support the claims made in the text.

11. Complete the results section to ensure all quantitative results are fully presented and aligned with figures and tables.

12. Include a discussion of negative results to provide a balanced view of the findings.
